

GRADES 7-10



EMERGENCY SUB PLANS PACK



for **DIGITAL TECHNOLOGIES**

★ Keep learning on track—even when plans change. ★

**VOLUME 2****COMPUTER
SYSTEMS**

CPU



RAM



STORAGE

TASK CARD

12

Write a program that flashes an LED on for 2 seconds and off for 2 seconds.

**NETWORKS**

SHORT ANSWER QUESTIONS



1. What is the main purpose of a router?

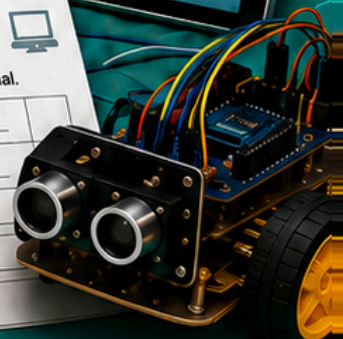
2. What is an IP address?

3. Give one example of a wireless network.

BINARY BASICS

Convert the following binary numbers to decimal.

1010	=	_____
1101	=	_____
1111	=	_____
10011	=	_____

**NO-PREP
ACTIVITIES****READY-TO-USE
RESOURCES****PERFECT FOR
EMERGENCY
LESSONS****GREAT FOR
SUBSTITUTE
TEACHERS****ENGAGING
DIGITAL TECH
TASKS****PRINT OR
DIGITAL USE****YOUR *LIFESAVER* IN THE CLASSROOM**

⌚ SAVE TIME

✔️ REDUCE STRESS

👥 KEEP STUDENTS ON TASK

**MADE BY TEACHERS
FOR TEACHERS**

Set this up while you are well

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
- 3 **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
- 4 **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
- 5 **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says “*but why?*”

This is Volume Two

Volume 2 stands completely on its own. Nothing in these ten lessons repeats Volume 1 — there is no binary conversion, no trace table, no flowchart and no password ranking here. If you own both, you have twenty substitute days that never overlap.

You do not need to know this subject

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 04.
- 2 **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
- 3 **Keep the teacher page and the answer key.** Do not photocopy those.
- 4 **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Clean the Data	Data errors and validation	★★	60	7-9
02	Charts That Lie	Data visualization and honesty	★	55	12-14
03	Databases by Hand	Records, fields, keys and queries	★★	60	17-19
04	Input, Process, Output	Systems thinking and sensors	★	55	22-24
05	Making It Smaller	Compression, lossy and lossless	★★	60	27-29
06	Secret Writing	Encryption, keys and key space	★★	60	32-34
07	Design for a Real User	User needs, criteria and interfaces	★	55	37-39
08	Break It Into Blocks	Functions, parameters and scope	★★★	60	42-44
09	Planning a Project	Task breakdown and scheduling	★	55	47-49
10	What a Free App Actually Costs	Data, consent and business models	★	55	52-54

Easiest to supervise

02, 04 and 10. Reading, judgment and argument. Students have opinions, so the room stays busy and nobody gets stuck.

Best for a quiet, working class

03, 05 and 08. Puzzle-like with definite answers. Strong students will race; that is what the extensions are for.

Best for a restless class

01, 07 and 09. Hunting for faults, sketching and shading a chart. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Cleaning data			
02 — Misleading charts			
03 — Databases			
04 — Systems and sensors			
05 — Compression			
06 — Encryption			
07 — Design and interfaces			
08 — Functions			
09 — Project planning			
10 — Free apps and data			

The first ten minutes

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is away. My name is _____. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Settle, hand out sheets	Use the script above
5-10	Read the opening page together	Ask one student to read it aloud
10-45	Students work through the tasks	Circulate. Check answers against the key
45-55	Extension task for those who finish	Point them to the last student page
55-60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 04 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

Clean the Data

2 MINUTES

Students find eight planted faults in a small dataset, write validation rules that would have stopped them, then answer questions the data can and cannot support. It is the least glamorous and most useful skill in the subject.

You do not need to know this subject

Every fault in the table is visible to anyone reading carefully — a missing number, an impossible date, a name typed twice. The key names all eight and explains each one.

Photocopy these pages only

Pages 7-9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A calculator is useful for one question but not essential.

WHAT TO SAY TO START

“Every organization you will ever work for has a database with junk in it. Today you are going to find the junk, work out how it got there, and write the rule that would have stopped it at the door.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0-5	Hand out sheets, settle	Use the door script on page 5
5-12	Read the panel and worked example together	Point out the four fault types
12-28	Task 1.1 — find the eight faults	There are exactly eight. Key lists them
28-40	Task 1.2 — write validation rules	Judgment, not arithmetic
40-50	Task 1.3 and 1.4 — using the cleaned data	One calculation. Key has the number
50-60	Task 1.5, collect sheets	Discussion prompt below

Questions students will ask

“Is a spelling mistake really an error?”

If it stops a computer matching two records, yes. ben carter and Ben Carter are different strings to a machine.

“Can I just delete the bad rows?”

Sometimes. But deleting Eve because her score is missing throws away everything else she has. That trade-off is Task 1.4.

“How many faults are there?”

Eight, in eight different rows. Row 10 is completely clean, which students find suspicious.

If they finish early

Task 1.5 asks how you would check 40,000 rows you cannot read. Strong students can then design a one-page entry form that makes seven of the eight faults impossible to type in the first place.

If the computers are working

Students can type the table into a spreadsheet and use conditional formatting or a filter to surface the faults. Ask them to find them on paper first, then see whether the machine agrees.

Curriculum focus · for the regular teacher’s records

Acquiring, validating and interpreting data; data types; the effect of data quality on the information produced. Discussion prompt for the last five minutes: *“A report says the Grade 9 average is 64. It was calculated from this table. Is the report wrong, or just unlucky?”*

Garbage in, garbage out

16 MIN

NAME

CLASS

DATE

Data is not information until somebody cleans it

A computer will happily add up a column that contains a score of 105, a blank, and the word **Nine**. It will produce a confident average and it will be wrong.

Real datasets arrive broken. Learning to see the breakage is the first job of anyone who works with data.

Missing

A field left empty. Not the same as a zero — a zero is a score, a blank is a question.

Impossible

A value outside what reality allows: a score of 105, a date of June 31.

Wrong type

Words where numbers belong. **Nine** cannot be added up.

Inconsistent

The same thing written two ways: 05/22/2025 and 05-22-2025.

Task 1.1 · Find the eight faults

DETECTIVE

Every row below except one has exactly one fault. Circle it, then write which of the four types it is. There is one extra type to watch for: the same person entered twice.

#	STUDENT	GRADE	SCORE	DATE JOINED	EMAIL	FAULT TYPE
1	Aisha Rahman	9	72	05/03/2025	aisha.r@school.edu	
2	ben carter	9	68	05/03/2025	ben.c@school.edu	
3	Chloe Nguyen	9	105	05/12/2025	chloe.n@school.edu	
4	Dan Oliveira	Nine	54	05/12/2025	dan.o@school.edu	
5	Aisha Rahman	9	72	05/03/2025	aisha.r@school.edu	
6	Eve Larsson	10		05/19/2025	eve.l@school.edu	
7	Farid Haddad	10	61	06/31/2025	farid.h@school.edu	
8	Gia Romano	10	88	05/22/2025	gia.r.school.edu	
9	Hana Kim	10	47	05-22-2025	hana.k@school.edu	
10	Isaac Bell	10	79	05/25/2025	isaac.b@school.edu	

Which row is clean? _____ Which fault would be hardest for a computer to notice on its own?

Task 1.1 · the eight faults

ROW	THE FAULT	TYPE
2	ben carter in lower case	Inconsistent format
3	Mark of 105	Impossible value
4	Grade recorded as Nine	Wrong type
5	Aisha Rahman entered a second time	Duplicate record
6	Score left blank	Missing value
7	06/31/2025 — June has 30 days	Impossible value
8	Email has no @	Malformed / wrong type
9	Date written with hyphens	Inconsistent format

Row 10 is clean. The hardest fault for a computer to catch is the duplicate: both copies are perfectly valid on their own. Nothing about row 5 is wrong except that row 1 already exists.

Task 1.2 · model rules

FIELD	RULE
Grade	Must be a whole number, 7 to 10
Score	Whole number, 0 to 100
Date	A real calendar date, mm/dd/yyyy, not in the future
Email	Exactly one @, a dot after it, no spaces
Student	No two records may share the same student ID

The hard one is the last. Every other rule can be checked by looking at one value. Finding a duplicate means comparing a value against every other record, and names are not unique anyway — which is why databases use an ID.

Task 1.4 · the duplicate

- Almost certainly the same person: same name reordered, same class, same school email pattern.
- Keep B — it was updated three months later, so the newer email is more likely to work. Copy nothing across; A holds nothing B lacks. Accept keeping A if the student argues the older address is the official one, provided they justify it.
- A **student ID**: one value, never reused, that belongs to a person rather than describing them. Accept 'unique ID', 'primary key' or 'student number'.

Task 1.3 · the numbers

- Eight usable records. Row 5 is deleted as a duplicate and row 3's score is unusable.
- Grade 9 scores left are 72, 68 and 54. $194 \div 3 = 64.67$. Accept 64.7 or 65 with working shown.
- Gia Romano**, 88.
- Four** — Farid, Gia, Hana, Isaac. Eve has a record but no score.
- No**. A zero says she took the task and scored nothing. A blank says nobody knows. Turning one into the other invents data and drags the average down. Exclude her from the average and say so.

Task 1.5 · checking 40,000 rows

Good answers name checks a machine can run: count the blanks in each column · sort each column and read the top and bottom values (outliers surface immediately) · check every value against its type · count how many times each ID appears · count the distinct spellings of a category.

The last word: a computer can test the *shape* of a value, not its truth. 68 is a perfectly valid score even if the student actually scored 86. Validation stops nonsense, not lies or typos that happen to be plausible.

Decoding, and choosing what to lose

Task 5.2 · Decode the hidden image

PUZZLE

The codes

Row 1: 16

Row 2: 3, 10, 3

Row 3: 4, 2, 4, 2, 4

Row 4: 16

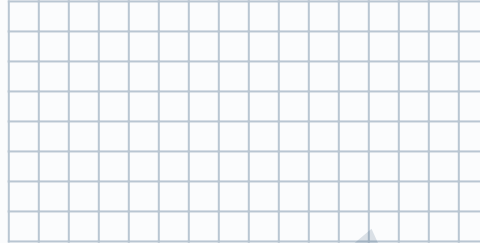
Row 5: 3, 1, 8, 1, 3

Row 6: 4, 8, 4

Row 7: 16

Row 8: 16

Shade the black runs. Remember the first number is always white.



What is it? _____

Task 5.3 · Which kind of smaller?

CLASSIFY

Lossless rebuilds the original exactly. **Lossy** throws information away permanently to save more space. Write L for lossless or Y for lossy, and say why it must be that one.

#	THE FILE	L / Y	WHY IT MUST BE THAT ONE
a	A photo posted to a social app		
b	A program you downloaded in a zip file		
c	A song streamed to your phone		
d	A hospital X-ray		
e	A spreadsheet of exam scores		

What do the lossless ones have in common? _____

Task 9.2 · Shade the schedule

DRAW

Shade one box per day of work, starting each task on the earliest day it is allowed to start.

TASK	1	2	3	4	5	6	7	8	9	10
A										
B										
C										
D										
E										
F										

- a. The video is finished at the end of day ____ b. Days spare before the deadline: ____
 c. Write down the chain of tasks that decides the finish date: _____

Task 9.3 · Now it rains

PREDICT

#	THE CHANGE	NEW FINISH DAY	WHY
a	Filming loses a day to rain (C takes 4)		
b	Permission forms take 3 days instead of 1		
c	Choosing the music takes 4 days instead of 1		
d	Editing is done in 2 days instead of 3		

Two of these four changed nothing at all. Those tasks are said to have **slack**. Which two, and roughly how much slack does each have? _____

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER

CLASS

DATE

PERIOD

Lesson used

CHECK ONE

- | | |
|--|--|
| ☐ 01 · Clean the Data | ☐ 06 · Secret Writing |
| ☐ 02 · Charts That Lie | ☐ 07 · Design for a Real User |
| ☐ 03 · Databases by Hand | ☐ 08 · Break It Into Blocks |
| ☐ 04 · Input, Process, Output | ☐ 09 · Planning a Project |
| ☐ 05 · Making It Smaller | ☐ 10 · What a Free App Actually Costs |

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher's day work, and it is genuinely appreciated.

Terms of use

Licensed for **one teacher in one classroom**. You may print and photocopy these pages for the students you personally teach, and you may pass the pack to a substitute teacher covering your own classes.

You may not redistribute it to colleagues, upload it to a school intranet or shared drive, resell it, or post it publicly. Additional licenses for a faculty or whole school are available at a discount.

© Digital Vector · digitalvector.com.au

Curriculum alignment

CSTA K-12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Acquiring, validating and interpreting data
- Representing data honestly in charts and visualizations
- Structuring data in related tables and querying it
- Digital systems: inputs, processes, outputs, sensors and actuators
- Compression and the representation of images
- Data security, encryption and keys
- User-centered design, criteria and evaluation
- Decomposition, modularity and reuse
- Project planning, dependencies and risk
- Privacy, consent and the economics of free software

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series

Volume 1 · Foundations

Algorithms, binary, trace tables, flowcharts, cyber attacks, networks, sorting, debugging, AI and digital footprints.

Volume 3 · Machines and Meaning

Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

Volume 4 · Media, Evidence and Impact

Images and sound as data, stacks and queues, digital forensics, version control, network design, accessibility, algorithmic decisions, IoT security and automation.

digitalvector.com.au

Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7-10

Volume Two · ten lessons · no preparation · no computers required